METRIC
FF-B-171/18
30 NOVEMBER 1993

FEDERAL SPECIFICATION

BEARING, BALL, ANNULAR, SINGLE ROW, PRIMARILY RADIAL, COUNTERBORED OUTER RING, CONTACT ANGLE UP TO AND INCLUDING 10°, NON-SEPARABLE, ANSI/AFBMA DIMENSION SERIES 03.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

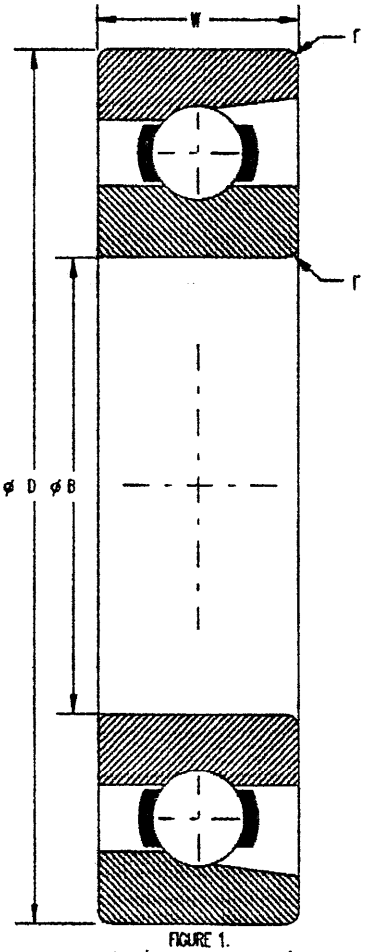


FIGURE 1.
OPEN (NO SHIELDS OR SEALS)
(CODE A)

BEARING CONFIGURATIONS

AMSC/NA

FSC 3110

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REQUIREMENTS:

1. Boundary dimensions. The boundary dimensions shall be specified in the part number by the size code in table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Shields and Seals. All primarily radial bearings with counterbored outer rings shall be open (no shields or seals, see figure 1) and designated by the code "A" in the part number.

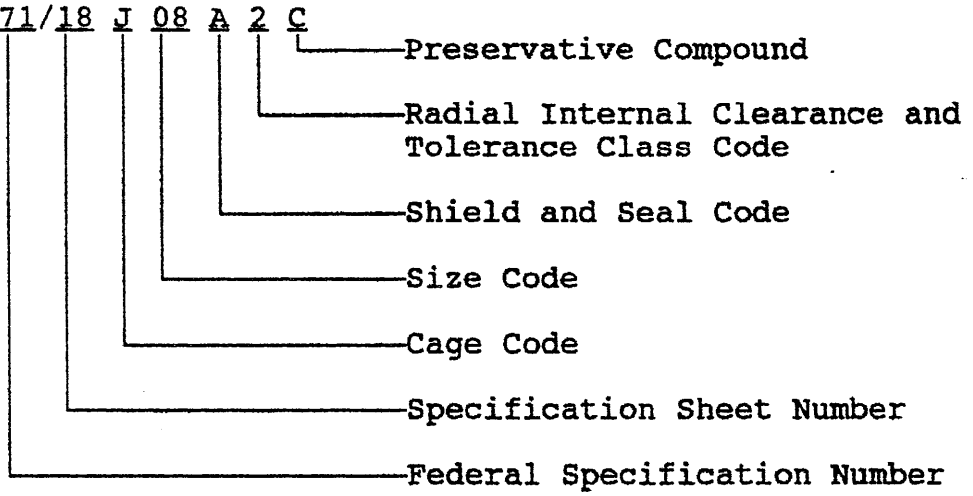
4. Radial Internal Clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table III.

5. Tolerance class. The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table III.

6. Preservative Compound. All bearings shall be supplied with preservative compound in accordance with MIL-P-197 and designated by the code "C" in the part number.

7. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, shields and seals, radial internal clearance and tolerance class, and preservative compound.

EXAMPLE: FFB171/18 J 08 A 2 C



FFB171/18J08A2C indicates - Bearing, cage-pressed steel or molded plastic; bore 40mm; outside diameter 90mm; width 23mm; open (no shields or seals); radial internal clearance symbol 0 (normal); tolerance class ABEC-1; perservative compound in accordance with MIL-P497.

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic Load Rating (Pounds)
01	12	37	12	1.0	18	1760
02	15	42	13	1.0	21	2005
03	17	47	14	1.0	23	2435
04	20	52	15	1.1	27	3325
05	25	62	17	1.1	32	3875
06	30	72	19	1.1	37	5120
07	35	80	21	1.5	43.5	6500
08	40	90	23	1.5	48.5	8040
09	45	100	25	1.5	53.5	9720
10	50	110	27	2.0	60	13360
11	55	120	29	2.0	65	15455
12	60	130	31	2.1	72	17680
13	65	140	33	2.1	77	19760
14	70	150	35	2.1	82	20190
15	75	160	37	2.1	87	25130
16	80	170	39	2.1	92	26040
17	85	180	41	3.0	99	29640
18	90	190	43	3.0	104	30435
19	95	200	45	3.0	109	32700
20	100	215	47	3.0	114	35055
21	105	225	49	3.0	119	37415
22	110	240	50	3.0	124	42175
24	120	260	55	3.0	134	47185
26	130	280	58	4.0	148	52350
28	140	300	62	4.0	158	57665
30	150	320	65	4.0	168	63575
32	160	340	68	4.0	178	69390
34	170	360	72	4.0	188	72450
36	180	380	75	4.0	198	78475
38	190	400	78	5.0	212	84640
40	200	420	80	5.0	222	90935
42	210	440	84	4.0	240	97360
44	220	460	88	5.0	242	103905

1/ Size codes in Table I are industry bore codes.

2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.

3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

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TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)
Y	Pressed brass

TABLE III. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (Less than normal)	ABEC-1
2	Symbol 0 (Normal)	
3	Symbol 3 (Greater than normal)	
4	Symbol 4 (Greater than symbol 3)	
5	Symbol 2 (Less than normal)	ABEC-5
6	Symbol 0 (Normal)	
7	Symbol 3 (Greater than normal)	
8	Symbol 4 (Greater than symbol 3)	

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force - 11

Review Activities:
Army - AR, EA, MI
Navy - OS

User Activity:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

PREPARING ACTIVITY:
DLA-IS

(Project 3110-0848-18)

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